Chap 6. Lagrange's Theorem

Lagrange's Theorem

If a group G is finite and H is a subgroup of G, then $|H|$ divides $|G|$.

So. eg if $|G| = 100$ then we <u>might</u> see subgroups H of G with orders.

$1, 2, 4, 5, 10, 20, 25, 50, 100$

but we certainly won't see subgroups of orders $3, 6, 7, 23, 78, \ldots$ etc

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We've already seen this in Chapter 4 on Cyclic groups

If $G = \langle a \rangle$ and $|G| = |\langle a \rangle| = n$

we saw that for any k, $H = \langle a^k \rangle$ is a subgroup G of order $|H| = \frac{n}{d}$

where $d = \gcd(k, n)$. Notice that $\frac{n}{d}$ is a divisor of n.

Sketch of proof of Lagrange's Theorem

Given G and a subgroup H we can build a partition of G, one of whos subsets is H itself, and each subset has the same order (number of elements).

$$\Rightarrow |G| = \begin{pmatrix}\text{number}\\ \text{of subsets}\end{pmatrix} \times |H|$$

So therefore $|H|$ divides $|G|$ ∎

<u>Cosets</u>. Let G be a group with a subgroup H.

A <u>left coset of H in G</u> is a subset

$$gH = \{ gh : h \in H \}$$

where $g \in G$, could call g the representative of the coset.

A right coset of H in G is a subset

$$Hg = \{ hg : h \in H \}.$$

<u>Ex6.1</u> $G = (\mathbb{Z}_6, +)$

let $H = \{0, 3\}$

so the left cosets are

$0 + H = \{0, 3\}$

$1 + H = \{1, 4\}$

$2 + H = \{2, 5\}$

$3 + H = \{3, 0\} = H = 0 + H$

$4 + H = \{4, 1\} = 1 + H$

$5 + H = \{5, 2\} = 2 + H.$

<u>Notice</u>:

- different reps. can generate same coset.
- the different reps. of a coset are exactly the element of the coset.
- $\{0, 3\}, \{1, 4\}, \{2, 5\}$ is a partition of $\mathbb{Z}_6$.

- $\mathbb{Z}_6$ is abelian so any left coset is the same as the corresponding right coset.

<u>Ex 6.2</u> Consider $G = S_3$

$|G| = 6$. $H = \langle (1\,2\,3) \rangle$

Let $H = \{(1), (1\,2\,3), (1\,3\,2)\}$

Left cosets are

$(1)H = (1\,2\,3)H = (1\,3\,2)H = H$.

$(1\,2)H = (1\,3)H = (2\,3)H =$

$\{(1\,2), (1\,3), (2\,3)\}$.

$(1\,2\,3)(1\,2\,3) = (1\,3\,2)$

In fact the right cosets agree

here in that

$H(1) = H(123) = H(132) = H$

$H(12) = H(13) = H(23) =$

$\{(12), (13), (23)\}$

producing the same partition.

But the subgroup $K = \{(1), (12)\}$ produces different partitions by its left cosets

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<u>Lemma 6.3</u> Different ways to express that two left cosets are the same

<u>Proof</u> ~~examples~~

$1 \Rightarrow 2$.

Assume $g_1H = g_2H$.

$x = (x^{-1})^{-1}$

<u>claim</u> $Hg_1^{-1} = Hg_2^{-1}$

Pf: Let $\boxed{x \in Hg_1^{-1}}$

$\Rightarrow \exists\, h \in H$ $\quad x = hg_1^{-1}$

$= ((hg_1^{-1})^{-1})^{-1}$

$= (g_1h^{-1})^{-1}$

$= (g_2k)^{-1}$, for some $k \in H$

(since $g_1H = g_2H$.)

$= (k^{-1}g_2^{-1})$

$\boxed{\in Hg_2^{-1}}$

Therefore $Hg_1^{-1} \subset Hg_2^{-1}$

Similarly can prove that $Hg_2^{-1} \subset Hg_1^{-1}$

Therefore $Hg_1^{-1} = Hg_2^{-1}$, as required.

Other proofs as exercises.

Continue at 12:05.

<u>Theorem 6.4</u> Let H be a subgroup of G. The <u>left</u> cosets of H in G, partition G.

<u>Proof</u>: We need to show that the cosets are disjoint, and cover the whole of G.

① <u>Claim</u> Different cosets are disjoint.

Let g_1H, g_2H be two cosets, we claim.

$$\underbrace{g_1H = g_2H}_{A} \quad \text{OR} \quad \underbrace{g_1H \cap g_2H = \phi}_{B}.$$

OR

<u>Pf</u> Let's prove $(\neg B) \Rightarrow A$.

Suppose $g_1H \cap g_2H \neq \phi$

$\Rightarrow \exists\, a \in g_1H \cap g_2H$

$\Rightarrow \exists\, h_1, h_2 \in H \quad a = g_1h_1 = g_2h_2$

$\Rightarrow \quad g_1 = g_2h_2h_1^{-1}$

$\Rightarrow \quad g_1 \in g_2H$

$g_1 = g_2h_2h_1^{-1}$, $g_2 = g_1h_1h_2^{-1}$

any g_1k, for $k \in H$
$= g_2h_2h_1^{-1}k$
$\in g_2H$

$\Rightarrow \quad g_1H = g_2H$, by lemma 6.3.

$\Rightarrow g_1H \subset g_2H$

$g_2H \subset g_1H$.

This proves ①.

② Claim: the cosets cover G.

<u>Proof</u> Notice that if $g \in G$.

$$g \in gH = \underbrace{\{gh : h \in H\}}_{\substack{\cap \\ ge \\ = g}} \quad e \in H$$

① & ② prove that the collection of left cosets of H partition G.

<u>NB</u>: Can show in a very similar way that the right cosets of H in G, also partition G.

<u>Definition</u> The number of <u>left</u> cosets of H in G is called the "index of H in G" written as $[G : H]$

<u>Theorem 6.8</u> The number of left cosets of H in G = the number of ~~right cosets of H in G~~

<u>Proof</u> Construct a bijective mapping

$\phi : \mathcal{L}_H \longrightarrow \mathcal{R}_H$.

where $\mathcal{L}_H$ = set of left cosets

$\mathcal{R}_H$ = " " right "

Define ϕ by

$$\phi(gH) = H\underline{g^{-1}}$$

<u>Firstly</u> this map ϕ is well defined by this formula. since by lemma 6.3 if $g_1H = g_2H$ then $Hg_1^{-1} = Hg_2^{-1}$

<u>Claim ϕ is injective</u>

Suppose $\phi(aH) = \phi(bH)$

$\Rightarrow Ha^{-1} = Hb^{-1}$

$\Rightarrow aH = bH$ by lemma 6.3

~~this proves injectivity~~

<u>Claim ϕ is surjective</u>

Let $Ha \in \mathcal{R}_H$.

Notice. $\phi(\underline{a^{-1}}H) = Ha$ $= H(a^{-1})^{-1}$

So $\phi : \mathcal{L}_H \longrightarrow \mathcal{R}_H$ is bijective

$\Rightarrow |\mathcal{L}_H| = |\mathcal{R}_H|$.

Proposition 6.9 All left cosets of H in G have the same size, namely $|H|$.

Proof. Let's construct a bijection

$$f : H \to gH.$$

defined by

$h \in H \quad f(h) = g\underline{h}$

Can easily prove that f is bijective.

1-1 since if $f(h) = f(k)$

$$\Rightarrow gh = gk$$
$$\Rightarrow h = k$$

onto since, take any $x \in gH$.

$x = gh$, for some $h \in H$.

but then clearly $f(h) = x$

Therefore f is a bijection

$$\Rightarrow |H| = |gH|$$

$\blacksquare$

Lagrange's Theorem proof

Put all previous results together to see that (counting $|G|$ from the partition by left cosets)

$$|G| = [G:H]\,|H|$$

i.e. $|H|$ divides $|G|$. ∎

Corollary 6.12

Suppose $|G| = p$, $\boxed{p \text{ a prime.}}$

Then G is cyclic.

Proof $p \geq 2$. So let a be any non-identity element of G.

Consider $\langle a \rangle$ $\boxed{|\langle a \rangle| \geq 2}$

and by Lagrange's theorem $|\langle a \rangle|$ divides p.

$$\Rightarrow |\langle a \rangle| = p.$$

$$\Rightarrow \langle a \rangle = G.$$

i.e. G is cyclic.